

## Motel Corporation Product Brief: Cipher Personal Computer

The Cipher is a single-board microcomputer incorporating two popular microprocessors, extensive memory, advanced video graphics capabilities, and compatibility with a large base of existing computer programs.

It is a complete electronics package, requiring the addition of standard input/output devices to become a full computing system.

As a first computer, it offers the user the ability to execute a large base of existing "CP/M-80" software now in widespread use, while allowing future advancement to 16-bit "CP/M-68K" software and graphics applications. The Cipher will execute existing programs 4 to 10 times faster than most other personal computers.

As a second computer for more experienced users, it retains the capabilities of present systems as a springboard to the emerging 16-bit technology. Standard input/output devices may be taken from existing systems and used with the Cipher, lowering overall system cost.

### Primary Hardware Features:

#### Two microprocessors

- Z80        8-bit processor for existing CP/M-80 software
- 68000     16-bit processor for CP/M-68K software

#### Advanced Graphics

- high-resolution video graphics suitable for computer-aided design applications.
- 1024 by 1024 line resolution, 1 microsecond pixel drawing time.
- intelligent graphics controller allows dynamic display modification without heavy computing burden on main processor.
- high-density text display suitable for word-processing applications.

## Extensive memory

- standard 256K bytes for both Z80 and 68000 programs and data.
- memory management technique permits Z80 to access 68000's entire 16-megabyte addressing range, provides compatibility with CP/M version 3.

## Single-board construction

- reduces cost, simplifies design.

## I/O capabilities

- on-board dual-density floppy disk interface, serial and parallel ports.

## Primary Software Features

### Designed for CP/M

- fully compatible with CP/M-80 versions 2.2 and 3.0, and new CP/M-68K software.
- Disk-file compatibility maintained between CP/M-80 and CP/M-68K.

### Software Processor Selection

- 68000 can switch control to Z80 and vice versa.
- simplifies adaptation of high-level software from Z80 to 68000: Non-portable assembler code can be executed by Z80 without change, while bulk of high-level software can be re-compiled for faster execution by 68000.

### Large CP/M-80 Software Base

- including WordStar, VisiCalc, dBASE-II, financial packages.
- CP/M-80 version 3 capability permits existing programs to execute 4 to 10 times faster than on most other personal computers.

## CP/M-68K compatibility

- newly-available software offers complete 68000 development package, including assembler and C language compiler.

## Directions for Future Product Development

- 1 megabyte of main program memory
- hard-disk interface
- concurrent Z80/68000 processing
- colour graphics, Telidon decoding software
- UNIX operating system
- other microprocessors (ie. 186, 286, 16000 devices)
- industry-compatible versions (plug-in to Multibus etc.)
- applications software (word and image processing, computer-aided design)
- multi-user capability
- network capability

## Notes

The 8-bit Zilog Z80 is one of the most widely-used microprocessors for personal and business microcomputers. Examples of other Z80 systems include Radio Shack TRS-80, Xerox 820 IPS, BMC if800, and NABU Mfg. business systems.

The 16-bit Motorola 68000 is a highly advanced, architecturally-elegant microprocessor, selected by Apple Computer Corp. for use in their "Lisa" computer, and by IBM for a new laboratory computing system.

CP/M-80 and CP/M-68K are "operating systems" developed and marketed by Digital Research, Inc. An operating system is a software package which "standardizes" a computer, enabling it to execute readily-available software without modification.

CP/M-80 version 2.2 is used in most 8-bit business computers and many personal computers. Its existence has led to the development of a huge base of Z80-compatible software for business, scientific and personal applications. The newer version 3.0 offers enhanced performance with this existing software, but compatible systems are not yet widely available.

The introduction of CP/M-68K in early 1983 should stimulate development and availability of 68000 software, starting immediately.